Development and validation of a complexity score to rank hospitalized patients at risk for preventable adverse drug events

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Purpose. The development of risk models for 16 preventable adverse drug events (pADEs) and their aggregation into the final complexity score (C-score) are described.

Methods. Using data from 2 tertiary care facilities, logistic regression models were constructed for the first 5 hospital days that admissions were at risk for each of 16 pADEs. The best model for each pADE was validated in 100 bootstrap samples. The C-score was then aggregated and predicted individual pADE risk as the probability to develop at least 1 pADE. Using the 100 bootstrap samples for each pADE, 100 C-scores for validation were generated.

Results. We utilized electronic health records (EHR) data from 65,518 admissions to UF Health Shands and 18,269 admissions to UF Health Jacksonville to develop risk models for 16 pADEs. Most models had very strong discriminant validity (C-statistic > 0.8), with the highest predicted decile representing about half of manifest pADEs. Among admissions in the highest C-score decile, about two thirds experienced at least 1 pADE (C-statistic, 0.838; 95% confidence interval, 0.838–0.839). C-score precision, defined as the percentage of patients consistently (i.e., at least 95 of 100 samples) ranked in the 90th percentile, was 80–84%.

Conclusion. The C-score was developed and validated for the identification of hospitalized patients at highest risk for pADEs. Aggregation of individual prediction models into a single score reduced its predictive power for most pADEs, compared with the individual risk models, but concentrated in the highest C-score decile a patient group more than two thirds of whom experienced at least 1 pADE.

Keywords: drug-related side effects and adverse reactions, electronic medical record, inpatients, patient safety, prediction model, risk scores

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Greater demand for better quality and safety at lower costs, widespread adoption of health information technology, and pharmacists' expanding roles in population-based drug therapy management have created a need to redesign the pharmacy practice model.[1] In response, ASHP launched the Pharmacy Practice Model Initiative, now termed the Practice Advancement Initiative, which emphasizes pharmacists' key role in medication therapy management (MTM).[2] Because most current staffing models in acute care settings do not allow comprehensive pharmacist assessment and management of all hospital admissions, successful implementation of this initiative requires that pharmacists identify and prioritize patients with the greatest need for MTM.

Traditionally, 2 approaches have been used to allocate clinical phar-

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KEY POINTS

- An automated electronic health record–based algorithm (complexity score, or C-score) was constructed to flag admissions at risk for preventable adverse drug events (pADEs).
- The C-score, a composite of prediction models for 16 pADEs, showed very strong discriminant validity and concentrated in its upper decile a patient group more than two thirds of whom experienced at least 1 pADE.
- Once validated in other facilities, the C-score may offer an effective algorithm to allocate pharmacy resources to patients at greatest need of medication therapy management services.

macy services in hospitals, including unit-based criteria (i.e., assignment to units with complex and often costly medication regimens) and process-based criteria (i.e., assignment to support a particular drug therapy process such as management of special orders or protocols or response to clinical alerts). Both approaches are narrow in focus, concentrating on specific subpopulations that may or may not have a higher risk for preventable adverse drug events (pADEs), a major focus of pharmacist MTM. Our literature search identified only 1 tool that used previously reported pADE risk factors (e.g., advanced age, exposure to high-risk drugs) to develop a more-comprehensive risk stratification algorithm, but its predictive validity in accurately identifying inpatients who will experience pADEs is unknown.[3]

Measurement of a targeted adverse outcome and development of a risk model to predict this outcome for prospective risk stratification of patients are not novel approaches in other clinical specialties. Intensive care practitioners are familiar with sequential organ failure assessment (SOFA) scores used to identify patients at high risk for death[4] and models to project cardiac arrest[5] or a prolonged need of intensive care.[6] Other automated models used in acute care settings predict the risk of readmission[7] or surgical complications,[8] but no validated tool is available that summarizes the risk for pADEs. However, the use of broad outcomes, such as mortality, for risk stratification of pharmacy services lacks the specificity to adverse events that are associated with drug therapy and thus are not sensitive to pharmacist intervention.

To optimize pharmacist allocation to patients with the greatest need, the ASHP Research and Education Foundation funded the development of a complexity score (C-score) that prospectively identifies patients at greatest risk for pADEs and thus with the greatest need for pharmacist MTM. This is the final report in a series of 3 describing the development and validation of the C-score.[9,10]

Methods

In order to determine the best target population for pharmacist interventions, we sought to identify patients with the greatest risk for adverse drug events (ADEs) that were frequently deemed preventable through pharmacist MTM. The selection of candidate pADEs for C-score development has been described previously and included a systematic literature review coupled with a stakeholder survey to evaluate the relevance of each pADE for hospital pharmacy practice.[9] Of the 21 candidate pADEs considered for inclusion in the C-score prediction models, 16 were operationalized for measurement and forwarded for prediction modeling. Some of the original pADEs were dropped for C-score development due to their inability to measure the outcome in electronic health records (EHRs), too small incidences to allow risk model development, and concerns about limited preventability and limited appropriateness for pharmacist MTM.[10] The entire project, including pADE selection, was guided by a technical expert panel representing clinical pharmacy specialists, health information technology experts, and researchers in quality measurement and epidemiology.

Study population and pADE measurement. For C-score development, we constructed a retrospective cohort of hospitalized patients using inpatient EHR data from the two largest University of Florida (UF) affiliated hospitals: UF Health Shands (852 beds) and UF Health Jacksonville (695 beds). Data were extracted for all adult patients admitted from January 1, 2012, through October 31, 2013 (UF Health Shands), and from March 1, 2013, through October 31, 2013 (UF Health Jacksonville). The EHR system provider for UF Health hospitals and clinics is EPIC (Verona, WI). Characteristics of the study population and pADE measurement, including exact operational definitions and respective incidence estimates, have been published.[10] In brief, we used a variety of EHR data elements to determine and time the onset of each pADE. Each PADE was measured during risk days defined by in- and exclusion criteria to focus measurement on patients who were at a given hospital day at risk for the pADE. For example, assignment of a patient to a risk day required, for some pADEs, exposure to a medication that may cause the adverse event. Some pADEs were permitted to recur (e.g., hypoglycemia), while the manifestation of others terminated further inclusion in the risk population (e.g., venous thromboembolic events).

Development of pADE risk models. We developed risk models for each pADE to maximize predictive power. Details on the development and validation of the individual risk models have been published.[11-17] In brief, we identified pADE risk factors using literature searches, drug monographs, and clinical expert interviews. Particular emphasis was given to risk factors that reflect quality deficits in drug therapy and thus opportunities for pharmacist intervention.

We ascertained risk factor information for each risk day and during a "look back" period of up to 1 year as necessary. Each risk factor was assigned a duration to ensure clinically relevant proximity to adverse event onset. For example, administration of antidiabetic agents may be considered a risk factor for up to 24 hours, while warfarin may continue to be a risk factor for several days. Some risk factors were fixed (e.g., blood glucose concentration on admission) while other were updated (e.g., minimum blood glucose concentration of preceding day).

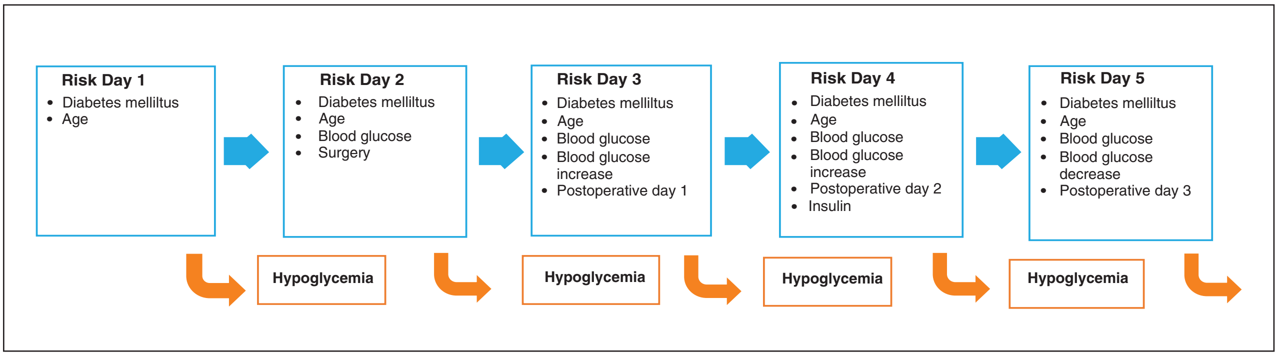
Each risk factor that was measurable from discrete fields in the hospitals' EHR system was operationalized and examined in descriptive and univariate analyses for plausibility and to determine its association with the target pADE. Clinical experts on our team validated risk factor–ascertainment algorithms via chart review and defined data-cleaning steps (e.g., remove extraneous values) as necessary. Risk factors with implausible frequencies or other features that suggested inaccurate measurement were discarded. We pursued 2 strategies to deal with missing values for risk factors: (1) imputation with normal or average values if only small proportions were missing and (2) use of a missing indicator variable if the fact that the information was not obtained was informative. To reduce the chance for model overfitting, we conducted cluster analyses to identify highly correlated risk factors and retained, among a given cluster, the risk factor with the strongest contribution to model fit (measured as the Akaike Information Criterion [AIC]). We further evaluated the predictive properties of various transformations of each continuous variable (e.g., log or cubic splines) in univariate analysis to arrive at a final set of candidate risk factor variables for modeling.

We ran separate risk models for each admission's first 5 risk days, with the models for later days accumulating more risk factor information. For example, while hospital day 1 can only consider information that is present on or before admission, hospital day 5 can consider all risk factor information that has been generated during the hospital stay (Figure 1). We used logistic regression models that predicted, for a given risk day, the manifestation of the pADE within the specified follow-up period. The total number of considered risk factors per model ranged from 21 to 52. Discharged patients were assumed to not have the pADE.

For each pADE, we constructed 4 models for each of the 5 risk days, including (1) a nonparsimonious model, (2) an expert model that included a set of risk factors considered most relevant by our clinical experts, (3) a parsimonious model that used fast backward elimination and retained all risk factors with an association that had a p value of <0.15, and (4) a reduced back-step model with risk factors that were retained at the same p value in at least 40 of 100 backward elimination processes using 100 bootstrap samples derived from the original study population. The best-fitting model for each risk day was then selected based on model fit (AIC), discriminative properties (congruence statistic [C-statistic]), and calibration (Hosmer-Lemeshow goodness of fit).[17]

Again, to reduce the risk for model overfitting, for pADEs with small inci-

Figure 1. Temporal organization of risk modeling for diabetes mellitus. In this example, diabetes mellitus is a fixed risk factor and blood glucose concentration is a dynamic risk factor in each risk day used to predict the preventable adverse drug event of hypoglycemia on the following day. During the hospital stay, information (e.g., changes in blood glucose concentration) improves, and the strength of risk factors (e.g., effect on blood glucose since surgery) may change.



dences, we explored whether the coefficients for risk factors changed significantly between the 5 risk models by fitting one model across all 5 risk days and examining interactions between risk factors and risk days. If no appreciable interactions were noted, we collapsed risk days by constructing a single model across several risk days by constructing a single model across several risk days to achieve more-precise model estimates. Risk factor definitions for each risk day were retained as originally defined for the individual day models (i.e., each risk day was treated as an independent observation). Because this created clusters of multiple observations that may be correlated, we examined the magnitude of the clustering effect on parameter estimates. To do so, we compared the parameter estimates from simple logistic regression assuming independence and those using generalized estimating equations accounting for the correlation. Since the results were close and the clustering had only marginal effects on parameter estimates, we used logistic regression for simplicity.

Finally, the selected model for each risk day was validated through replication in 100 bootstrap samples derived from the original study population. The original model is considered valid if its C-statistic is located within the 95% confidence interval (CI) of the C-statistics derived from the 100 bootstrap repetitions.[18] All final models that were included in the C-score met this validity criterion. Figure 2 illustrates pADE model development and validation and subsequent aggregation of the models in the C-score.

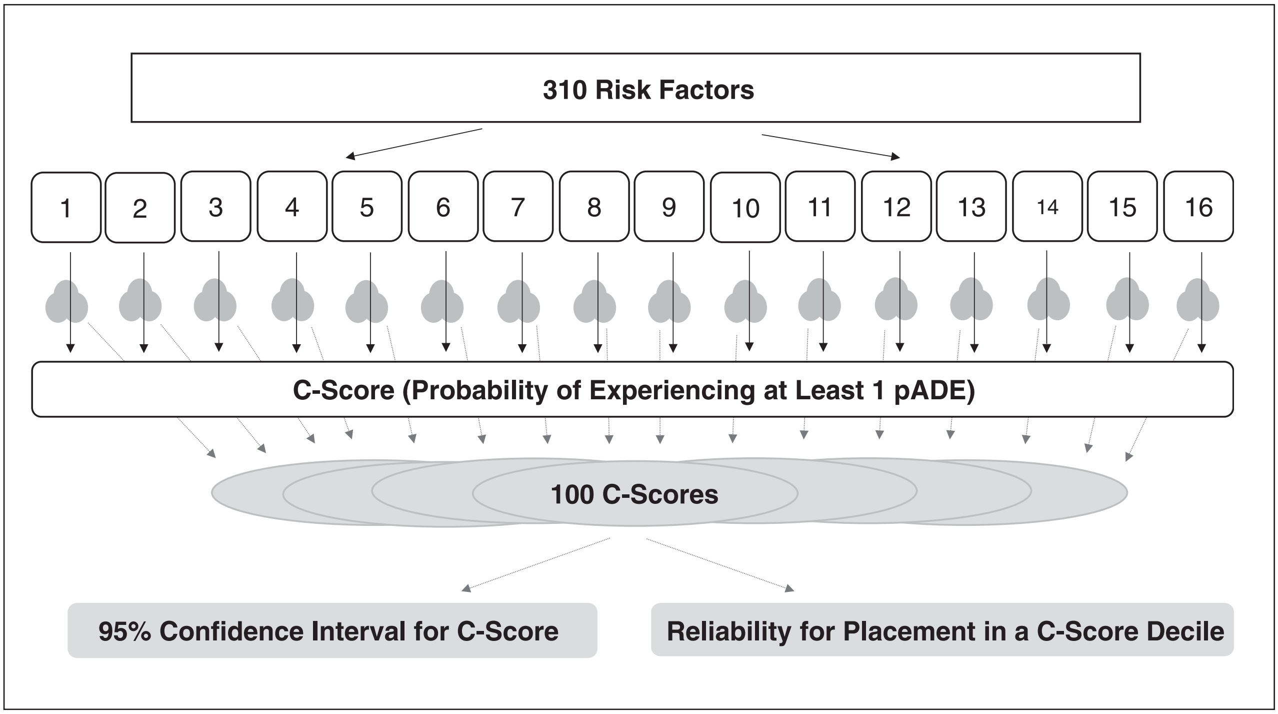
C-score assembly. The final C-score is defined as the predicted probability to experience at least 1 pADE after a given risk day. To compute the C-score for each patient, we calculated the probabilities for the manifestation of each pADE during the specified follow-up period for each of the first 5 hospital days. If a patient was not in the risk pool for a given pADE at a given hospital day, his or her probability for this pADE was considered 0. Assuming independence of pADEs, a patient's C-score was the aggregate of all 16 probabilities for a given hospital day, as:

$$\text{C-score} = 1 - \prod (1 - p_{\text{pADE}})$$

where p_{pADE}= probability of a given pADE at a given hospital day.

To assess C-score uncertainty, which is inherent in the uncertainty of the estimates of 16 pADE models, we used the original results from the 100 bootstrap models computed for each pADE model's validation. Specifically, we used the parameters estimated by

Figure 2. Schematic flow of C-score development (shapes with black frames) and validation (gray shapes). A total of 310 risk factors were incorporated into models for 16 preventable adverse drug events (pADEs). Each pADE model was validated using 100 bootstrap samples. To formulate a single C-score, pADE probabilities were summarized as the probability to have at least one pADE. The original 100 bootstrap samples for each pADE were then used to validate the C-score and obtain 95% confidence intervals.



each pADE's 100 bootstrap validation models to calculate 100 probabilities for each of the 16 pADEs for each patient. The resulting matrix of 1,600 probabilities for each patient-day was used to calculate 100 C-scores via random sampling of 1 of the 100 probabilities for each pADE. These were then used to calculate 100 C-statistics to derive CIs and estimate the proportion of times the same patient was assigned to the same C-score decile. Agreement between the predicted and observed risks for at least 1 pADE calculated for each patient at each of the first 5 hospital days was also plotted for each C-score decile.

We conducted all analyses in SAS, version 9.4 (SAS Institute, Cary, NC), and used R (R Foundation for Statistical Computing, Vienna, Austria) to develop graphs. This study was approved by the UF institutional review and privacy boards.

Results

Our study population included 42,593 patients and 65,518 admissions in UF Health Shands and 13,314 patients and 18,269 admissions in UF Health Jacksonville. Across all 16 pADEs, patients had a total of 231,703 and 56,620 unique risk days that were used for modeling in UF Health Shands and UF Health Jacksonville, respectively. Several pADE models were collapsed across risk days (e.g., drug-associated falls with only 1 risk model for all 5 hospital days, drug-associated hypoglycemia with separate risk models for hospital days 1 and 2 and a collapsed model for hospital days 3–5) (Table 1). The number of patients at risk for pADEs varied and typically decreased during later hospital days because of increasing discharge rates. Smaller risk populations were noted for pADEs that required previous exposure to certain high-risk drugs (e.g., drug-induced hypoglycemia requires exposure to antidiabetic agents at the risk day) and those capturing certain uncontrolled conditions (e.g., uncontrolled pneumonia requires a prior diagnosis of pneumonia).

Incidences of pADEs (i.e., the number of admissions included at a given risk day in which patients developed the pADEs during the designated follow-up time) ranged from less than 0.5% for drug-associated acute mental status changes and venous thromboembolism (VTE) to more than 30% for uncontrolled pneumonia. The largest numerical contributors to pADEs overall were drug-induced hypotension, hyperglycemia, uncontrolled hypertension, and uncontrolled postsurgical pain, each affecting more than 3% of the entire study population.

The pADE models had excellent predictive properties, most with C-statistics above 0.8 (Table 1). The models with the lowest and highest C-statistics were those developed to predict drug-induced falls (0.686) and hyperglycemia (0.953, day 2 model), respectively. When focusing on patients ranked in the upper 50th percentile of the risk score, most models captured at least 80% of all patients on a given risk day who experienced the ADE during follow-up. The best performing model, hyperglycemia, included nearly all patients (>98%) with hyperglycemia in the upper half of the resulting score (Table 1). If only 10% of the risk population could be targeted with MTM services (i.e., patients in the 90th percentile of the C-score), the majority of models captured about half of all patients with the actual pADEs.

To simulate application of the C-score in real practice, we calculated the C-score for the first 5 hospital days of our study population, including totals of 215,637 and 56,620 days and 48,474 (22.5%) and 9,638 (17.0%) days with pADEs in UF Health Shands and UF Health Jacksonville, respectively. About 2.9% of risk days had more than 1 pADE.

Figure 3 shows the actual incidence of at least 1 pADE by C-score deciles. Of admissions ranked in the upper half of the C-score, 33.5–38.4% saw the development of at least 1 pADE, depending on the hospital day. About two thirds of admissions in the highest decile of the C-score were associated with a pADE (63.3% at hospital day 1 and 69.6% at hospital day 5, respectively).

Table 2 illustrates the incidences of the 16 pADEs among admissions in the highest C-score decile. Consistent with respective incidences, pADEs that were frequently captured by the C-score included hypotension, hyperglycemia, hypertension, and uncontrolled pain.

Table 3 reports the percentage of the total number of each of the 16 pADEs that was observed in admissions in the upper C-score decile. Among all admissions during which a pADE developed, the proportion captured in the 90th percentile of the C-score varied across risk days from 11% to 20% for VTE and from 48% to 84% for uncontrolled *Clostridium difficile* infection.

The precision of the C-score rankings, defined as the percentage of admissions that were consistently (i.e., ≥95 of 100 bootstrap samples) ranked in a given C-score percentile, was about 93% for placement in the upper half of the C-score and between 80 and 84% for placement in the 90th percentile (Table 4).

The pADE probabilities estimated by the C-score were largely consistent with observed pADE rates (Figure 4). Observed rates were closely scattered around the C-score values across all deciles of the score. Observed and predicted pADE rates are shown in the appendix, suggesting good calibration, though several pairs had nonoverlapping CIs, indicating significant differences between predicted and observed probabilities. The C-statistic for the C-score was 0.838 (95% CI, 0.838–0.839).

Discussion

To our knowledge, this study is the first attempt to develop an automated EHR-based prediction algorithm that allows ranking of patients according to their aggregate risk for ADEs and thus greatest need for pharmacist MTM. The C-score summarizes the

Table 1. Frequency of pADEs and C-Score Model Performance[a]

pADE and Risk Day(s)	No. Admissions in Risk Population	No. (%) Admissions With pADE	C-Statistic	No. (%) pADEs in Upper 50th Percentile of C-Score	No. (%) pADEs in Upper 90th Percentile of C-Score
Drug-induced acute mental status changes					
1–5	66,875	262 (0.39)	0.854	242 (92.4)	159 (60.7)
Drug-associated acute kidney injury[b]					
1	45,235	630 (1.39)	0.794	557 (88.4)	287 (45.6)
2	50,050	762 (1.52)	0.813	672 (88.2)	384 (50.4)
3	39,260	684 (1.74)	0.809	606 (88.6)	346 (50.6)
4	29,646	574 (1.94)	0.793	504 (87.8)	281 (49.0)
5	23,450	484 (2.06)	0.800	426 (88.0)	231 (47.7)
Drug-associated hypoglycemia					
1	12,264	294 (2.40)	0.831	277 (94.2)	140 (47.6)
2	15,810	358 (2.26)	0.860	337 (94.1)	212 (59.2)
3–5	32,688	604 (1.84)	0.877	579 (95.9)	381 (63.0)
Drug-associated hypotension					
1	28,508	3,941 (13.82)	0.725	3,004 (76.2)	1,177 (29.9)
2	31,855	3,360 (10.55)	0.797	2,876 (85.6)	1,310 (39.0)
3–5	63,961	6,093 (9.53)	0.809	5,250 (86.2)	2,605 (42.8)
Drug-associated acute liver injury[b]					
1	52,709	400 (0.76)	0.757	332 (83.0)	193 (48.3)
2	56,316	363 (0.64)	0.767	298 (82.1)	174 (47.9)
3	43,673	275 (0.63)	0.742	229 (83.3)	122 (44.4)
4	33,036	245 (0.74)	0.737	200 (81.6)	111 (45.3)
5	26,106	216 (0.83)	0.718	171 (79.2)	92 (42.6)
Drug-associated falls					
1–5	75,036	466 (0.62)	0.686	364 (78.1)	144 (30.9)
Uncontrolled hyperglycemia					
1	55,907	2,955 (5.29)	0.938	2,897 (98.0)	2,281 (77.2)
2	46,854	2,328 (4.97)	0.953	2,306 (99.1)	1,925 (82.7)
3	35,740	1,851 (5.18)	0.950	1,835 (99.1)	1,483 (80.1)
4	27,180	1,464 (5.39)	0.947	1,445 (98.7)	1,170 (79.2)
5	22,812	1,221 (5.60)	0.940	1,203 (98.5)	909 (74.4)
Hypokalemia					
1	82,137	1,539 (1.87)	0.830	1,397 (90.8)	835 (54.3)
2	66,365	1,087 (1.64)	0.846	1,012 (93.1)	608 (55.9)
3	49,501	814 (1.64)	0.849	751 (92.2)	492 (60.4)
4–5	65,433	1,071 (1.64)	0.840	974 (90.9)	618 (57.7)

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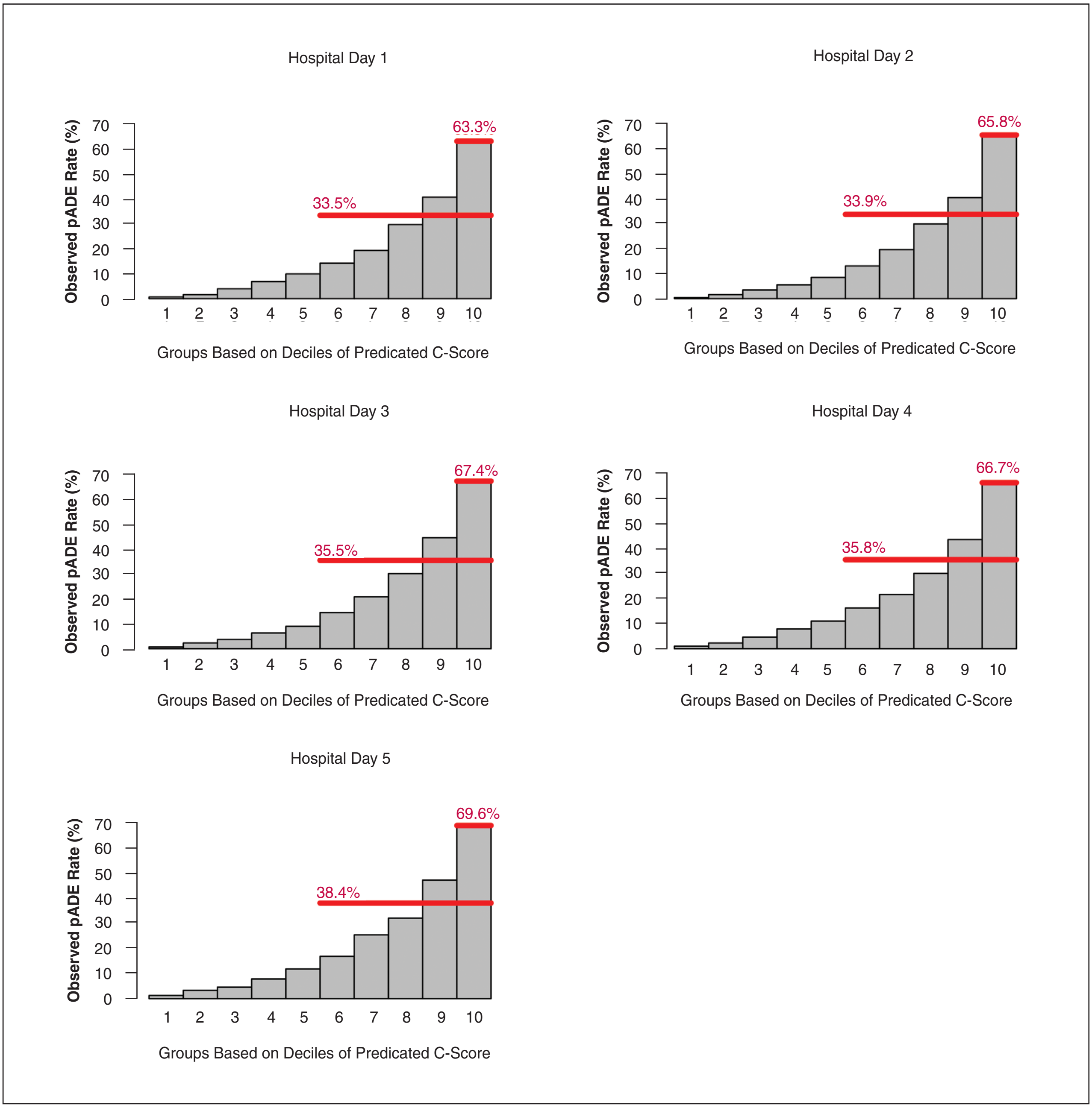
Table 1. Frequency of pADEs and C-Score Model Performance[a]

pADE and Risk Day(s)	No. Admissions in Risk Population	No. (%) Admissions With pADE	C-Statistic	No. (%) pADEs in Upper 50th Percentile of C-Score	No. (%) pADEs in Upper 90th Percentile of C-Score
Hyperkalemia					
1	81,857	439 (0.54)	0.919	421 (95.9)	354 (80.6)
2–5	180,457	418 (0.23)	0.896	395 (94.5)	320 (76.6)
Hyponatremia					
1–5	79,169	2,101 (0.83)	0.886	1,992 (94.8)	1,492 (71.0)
Drug-associated Q-T-interval prolongation					
1	43,850	799 (1.82)	0.825	755 (94.5)	351 (43.9)
2–5	121,997	873 (0.72)	0.803	804 (92.1)	362 (41.4)
Uncontrolled *Clostridium difficile* infection					
1–5	675	176 (26.07)	0.843	658 (87.9)	216 (28.8)
Uncontrolled hypertension					
1	83,787	3,747 (4.47)	0.905	3,611 (96.4)	2,319 (61.9)
2	68,456	2,479 (3.62)	0.917	2,418 (97.5)	1,660 (67.0)
3–5	120,014	4,654 (3.88)	0.921	4,548 (97.7)	3,080 (66.2)
Uncontrolled pneumonia[b]					
1	1,151	389 (33.80)	0.749	304 (78.1)	82 (21.1)
2	1,468	527 (35.90)	0.824	443 (84.1)	121 (23.0)
3	1,354	527 (38.92)	0.867	453 (86.0)	126 (23.9)
Uncontrolled postsurgical pain					
1	19,640	4,533 (23.08)	0.801	3,755 (82.8)	1,367 (30.2)
2	16,584	3,076 (18.55)	0.839	2,719 (88.4)	1,063 (34.6)
3	13,240	2,368 (17.89)	0.860	2,132 (90.0)	935 (39.5)
4–5	18,253	3,507 (19.21)	0.871	3,167 (90.3)	1,336 (38.1)
Venous thromboembolism[b]					
1	74,283	210 (0.28)	0.791	187 (89.0)	94 (44.7)
2	59,971	188 (0.31)	0.819	174 (92.6)	94 (50.0)
3	46,632	171 (0.37)	0.795	153 (89.5)	75 (43.9)
4	35,972	154 (0.43)	0.794	138 (89.6)	72 (46.8)
5	28,324	148 (0.52)	0.772	127 (85.8)	64 (43.2)

[a]pADE = preventable adverse drug event, C-score = complexity score.

[b]Estimates capture some cases multiple times because the follow-up time after the risk modeling day exceeds 1 day. For example, for drug-associated acute kidney injury, a follow-up time of 5 days after the risk modeling day is used to capture delayed drug effects. Thus, the onset of acute kidney injury on hospital day 4 would be attributable to risk modeling days 1– 3.

Figure 3. Observed rates of preventable adverse drug events (pADEs) per C-score decile on hospital days 1–5. The horizontal red lines represent the average observed pADE rate in hospital admissions in the upper half and the highest decile of the C-score.



risk for 16 ADEs that have been frequently reported as preventable and that are considered important targets for pharmacist intervention.

C-score development successfully addressed 2 unique challenges: the composite nature of the outcome and changes in risk as patients progress during admission. Considering the broad range of pADEs and their respective etiologies, it was expected that risk factors and their strength of association with each of the pADEs would vary. In fact, a given covariate may be a strong cause of 1 pADE and protective of another. Therefore, each pADE was modeled separately and then consolidated into an aggregate score.

The strength of prediction models depends greatly on the availability of measurable predictors of the outcome (risk factors) as well as the time when these variables are measured. Simply

Table 2. Observed pADEs per Hospital Day in Patients in Upper C-Score Decile[a]

pADE	No. (%) Admissions With pADE on Indicated Hospital Day				
	1 (*n* = 8,379)	2 (*n* = 6,846)	3 (*n* = 5,144)	4 (*n* = 3,850)	5 (*n* = 3,009)
Drug-associated acute mental status changes	23 (0.3)	13 (0.2)	<10	<10	<10
Drug-associated acute kidney injury	207 (2.5)	222 (3.2)	188 (3.7)	153 (4.0)	110 (3.7)
Drug-associated hypoglycemia	123 (1.5)	134 (2.0)	89 (1.7)	44 (1.1)	51 (1.7)
Drug-associated hypotension	1,384 (16.5)	1,130 (16.5)	791 (15.4)	563 (14.6)	426 (14.2)
Drug-associated acute liver injury	79 (0.9)	89 (1.3)	62 (1.2)	46 (1.2)	34 (1.1)
Drug-associated falls	<10	14 (0.2)	11 (0.2)	10 (0.3)	<10
Uncontrolled hyperglycemia	1,841 (22.0)	1,439 (21.0)	1,101 (21.4)	833 (21.6)	615 (20.4)
Hypokalemia	313 (3.7)	203 (3.0)	132 (2.6)	75 (1.9)	51 (1.7)
Hyperkalemia	129 (1.5)	51 (0.7)	27 (0.5)	16 (0.4)	19 (0.6)
Hyponatremia	86 (1.0)	124 (1.8)	121 (2.4)	96 (2.5)	100 (3.3)
Drug-associated Q-T-interval prolongation	188 (2.2)	79 (1.2)	46 (0.9)	25 (0.6)	15 (0.5)
Uncontrolled *Clostridium difficile* infection	16 (0.2)	27 (0.4)	34 (0.7)	31 (0.8)	40 (1.3)
Uncontrolled hypertension	950 (11.3)	479 (7.0)	367 (7.1)	278 (7.2)	187 (6.2)
Uncontrolled pneumonia	312 (3.7)	437 (6.4)	444 (8.6)	NA	NA
Uncontrolled postsurgical pain	1,049 (12.5)	1,239 (18.1)	1,027 (20.0)	1,019 (26.5)	945 (31.4)
Venous thromboembolism	35 (0.4)	39 (0.6)	29 (0.6)	17 (0.4)	16 (0.5)

[a]Counts of <10 are suppressed to protect privacy of personal health information. Hospital day denotes the day the C-score was calculated (i.e., the risk day); manifestation of pADEs occurred during the specified follow-up periods; no pADEs were measured on hospital day 1. pADE = preventable adverse drug event, C-score = complexity score, NA = not applicable.

focusing on patient characteristics that are present on admission may show weak associations with pADEs that develop later as a patient progresses during admission. As the diagnostic workup takes time, information about a patient's condition at admission may also take several days to be complete. These factors necessitated the development of dynamic prediction models that appropriately capture temporal and proximate relationships between preexisting and emerging risk factors and pADEs. Using these carefully developed definitions of risk days, the duration of follow-up to determine pADE manifestation, and the duration of each risk factor's contribution to risk, we developed separate models for the first 5 risk days. The individual pADE prediction models showed excellent performance, with C-statistics above 0.8 for most. For comparison, prostate-specific antigen predicts prostate cancer with a C-statistic of 0.84, SOFA scores predict intensive care unit mortality with a C-statistic range of 0.61–0.88, and risk scores for national readmission measures used in pay-for-performance agreements by the Centers for Medicare and Medicaid Services have C-statistics below 0.7.[4,19,20]

Not surprisingly, the C-score, which aggregates the estimated probabilities for each pADE into the probability to experience at least 1 pADE, performed equally well, with a C-statistic of 0.84. Comparisons between the predicted and observed risks for at least 1 pADE per C-score decile showed excellent congruence, and the C-score's precision in placing a given patient in the highest risk decile was consistently above 80%.

From a clinical perspective, focusing on patients in the highest C-score decile (i.e., only 10% of all adult admissions) would narrow pharmacy interventions to a high-risk population, more than two thirds of whom will experience a pADE. These pADEs will largely include drug-associated hypotension, uncontrolled hypertension, uncontrolled hyperglycemia, and uncontrolled postsurgical pain, which had the highest overall frequency in

Table 3. Percentage of Observed pADEs Captured on Admission in the Upper C-Score Decile[a]

pADE	% pADEs on Indicated Hospital Day 1	2	3	4	5
Drug-associated acute mental status changes	31.1	22.0	. . .[b]	. . .	. . .
Drug-associated acute kidney injury	32.9	29.1	27.5	26.7	22.7
Drug-associated hypoglycemia	41.8	37.4	33.7	24.9	31.3
Drug-associated hypotension	35.1	33.6	31.3	29.0	26.2
Drug-associated acute liver injury	19.8	24.5	22.5	18.8	15.7
Drug-associated falls	. . .	12.2	11.0	12.5	NA
Uncontrolled hyperglycemia	62.3	61.8	59.5	56.9	50.4
Hypokalemia	20.3	18.7	16.2	12.0	11.5
Hyperkalemia	29.4	29.7	25.2	22.9	27.5
Hyponatremia	16.8	26.1	27.6	28.0	30.2
Drug-associated Q-T-interval prolongation	23.5	21.0	20.3	16.3	12.9
Uncontrolled *Clostridium difficile* infection	84.2	52.9	57.6	48.4	54.1
Uncontrolled hypertension	25.4	19.3	18.6	18.4	15.9
Uncontrolled pneumonia	80.2	82.9	84.3	NA	NA
Uncontrolled postsurgical pain	41.5	51.6	51.9	59.0	60.0
Venous thromboembolism	16.7	20.7	17.0	11.0	10.8

[a]pADE = preventable adverse drug event, C-score = complexity score, NA = not applicable.
[b]Suppressed due to numerator count of <10.

our study population. Most pADEs that occur with lesser frequency were also well represented: focusing on admissions in the highest C-score decile would capture 50–80% of all patients who will develop uncontrolled *C. difficile* infection and 25–40% of all patients who will develop hypoglycemia. However, admissions in the highest C-score decile represented only between 10 and 20% of patients who will experience VTE and about 12% of patients who will experience drug-associated falls, not much more than would be expected from a random sample of admissions. Such differences in representation occur because of differences in the predictive validity of the individual pADE models (the individual risk models for falls and VTE have lower C-statistics) and correlations between pADE risks. In general, because patients at high risk for an individual pADE essentially compete for inclusion in the highest C-score decile, the C-score's ability to capture each pADE risk is expected to be inferior to the ability of the respective individual risk model. Indeed, comparisons between the percentage of admissions during which pADEs occurred that would be captured in the highest decile of the individual risk models versus the C-score show that the individual models tend to be superior in filtering the relevant highest-risk patients. Exceptions are uncontrolled *C. difficile* infections, uncontrolled pneumonia, and uncontrolled postsurgical pain, which are better captured in the highest decile of the C-score than via their corresponding individual models.

The observation that individual pADE models tend to be superior to an aggregate score for all pADEs illustrates the balance between specialization and broad care approaches that needs to be considered in directing pharmacists' attention. The C-score allows a focus on a broad spectrum of admissions with high likelihood of developing at least 1 pADE. In contrast, use of individual risk prediction models will better capture patients who will experience a particular pADE; however, considering the upper decile of each of the 16 pADE prediction models will also increase the number of patients who would need to be targeted. Thus, the C-score may be an alternative measure if pharmacy resources are limited. Importantly, because the C-score comprises individual pADE risk models, it can be deconstructed into its individual components. This implies that certain pADEs can be removed or weighted to emphasize or de-emphasize the C-score's focus. Such flexibility may be important to accommodate pharmacy capacity or institutional priorities.

The C-score does have several limitations. First and most importantly, while our internal validation has demonstrated strong validity, the C-score should be validated in other institutions to ensure accurate and

Table 4. Uncertainty Estimates for Placement in C-Score Deciles[a]

Hospital Day	% Admissions Ranked in Indicated C-Score Percentile in ≥95 of 100 Bootstrap Samples	
	50th C-Score Percentile	90th C-Score Percentile
1	94.0	80.4
2	93.4	82.4
3	93.2	83.3
4	93.0	83.7
5	93.0	83.2

[a]C-score = complexity score.

precise prediction of patients at greatest risk for a pADE. Several aspects that contribute to validity should be considered in this regard. The C-score uses 310 risk factors that are generated from 135 data elements from our institutions' EHR. To optimize predictive validity, we considered a broad spectrum of EHR elements, including not only laboratory values or medication administration records but also orders and clinical flow-sheet entries. While some of these elements are increasingly standardized across institutions, others depend significantly on local care processes and documentation practices. For example, conventions to document patients' mental status, stool appearance, or even pain may vary across institutions, and different scales and value sets may have different meanings and different predictive properties. Furthermore, healthcare practices that generate EHR data may vary, and the simple presence or absence of a test or procedure may have different meaning and different predictive properties. For example, our models found that missing albumin tests were often protective of pADE onset, suggesting that these tests were not ordered when patients were stable. Such protective features would vanish in institutions where albumin tests are more commonly part of a standard blood panel. Likewise, in testing the contributions of drugs to pADE occurrence, we were confined to the medications on our institutions' formulary, so predictive properties of drug groups or drug classes are not generalizable. Thus, local documentation practices and care delivery itself are expected to affect risk factor definitions and their contribution to pADE prediction. Institutions seeking to implement the C-score should therefore replicate our prediction models in their own EHR data system, which will provide institution-specific parameter estimates for pADE risk (i.e., C-score) calculations. Once implemented, institutions should have a maintenance plan that reevaluates the C-score when new drugs enter the market and there are other changes in healthcare delivery, as these may affect risk factor definitions and related parameter estimates.

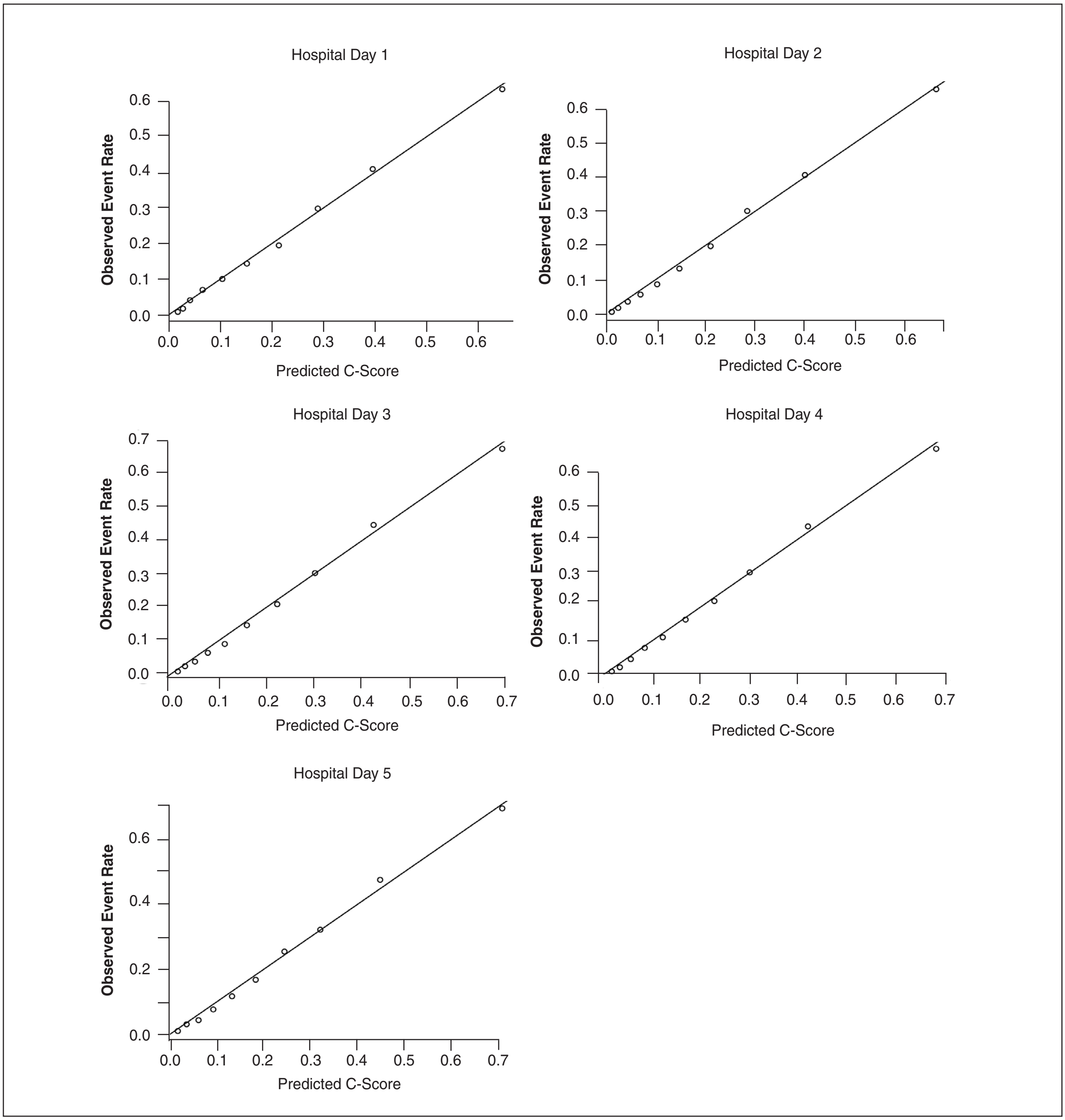
Second, in defining pADEs we aimed to ensure a strong contribution of preventable drug-related causes to the adverse event, but at least a fraction of captured events is expected to have different etiology and may be insensitive to pharmacists' (or anyone's) intervention. For example, in defining pADEs secondary to drug overdoses, we required prior exposure to these drugs. Thus, while our use of the term *preventable* is somewhat liberal, we expect that the majority of flagged patients will present with opportunities for pharmacist interventions. An exclusive focus on adverse events that are truly caused by inappropriate drug therapy in prediction modeling would require expert review of each case's etiology, including an assessment of causality and preventability, which has its own validity issues.

It is also important to note that the C-score does not predict which patients will benefit from pharmacist MTM services. Even under the assumption that only pADEs are included in the prediction, pharmacists' effectiveness in preventing these pADEs needs to be assessed via prospective C-score implementation. Because all pADE measures are automated, tracking of incidents to evaluate effectiveness is feasible and would provide the ultimate assessment of the C-score's utility in supporting pharmacist MTM services.

Third, we confined development of our prediction models to the first 5 risk days because sample size became too small to derive stable parameter estimates thereafter. We did note that estimated associations between risk factors and the pADE tended to be similar in later hospital days, which allowed for the collapsing risk days into single risk models in some cases and enhanced model precision. Based on this observation, we suggest that the last risk model (e.g., the model developed for risk day 5) can be applied to later hospital days using the same look-back period to ascertain risk factors that were used for the original model (e.g., 5 days), but further testing with larger samples would be ideal before pursuing such an approach.

Finally, the C-score currently predicts the joint probability of 16 pADEs. The selected pADEs originated from a comprehensive literature search and were validated in terms of their clinical relevance via stakeholder survey. In selecting thresholds in our pADE definitions, we aimed to define serious events that our expert panels agreed should not occur during admission and thus presented appropriate targets for pharmacist intervention. Unfortunately, we had to cease the consideration of several pADEs because their incidences were too low to allow development of stable pre-

Figure 4. Relationship between C-scores (the predicted probabilities of a predictable adverse drug event [pADE] C-score) and observed probabilities for at least 1 pADE.



diction models or because the events could not be reliably ascertained via automated EHR extraction.[10] Likewise, some risk factors that rely on narrative information in progress notes (i.e., not available in discrete fields) or genetic information that is not routinely collected may be readily accessible as EHRs advance and may allow further improvement of prediction models. Future research in larger hospital systems and using more sophisticated data ascertainment algorithms should focus on C-score expansion.

Conclusion

The C-score was developed and validated for the identification of hospitalized patients at highest risk for pADEs. Aggregation of individual prediction models into a single score reduced its predictive power for most pADEs, compared with the individual risk models, but concentrated in the highest C-score decile a patient group more than two thirds of whom experienced at least 1 pADE. The C-score algorithm—including EHR data retrival, validation, and prediction modeling—should be replicated in a user's own data system before implementation.

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Appendix—Observed and predicted pADE incidences by decile of the C-score[a]

Decile, by Day	*n*	Mean Observed pADE Incidence (%, 95 CI)	Mean Predicted pADE Incidence (C-Score, 95% CI)
Day 1			
1	8,378	0.71 (0.63–0.91)	1.63 (1.53–2.20)
2	8,379	1.66 (1.56–1.78)	2.67 (2.52–3.29)
3	8,379	3.98 (3.74–4.18)	4.04 (3.86–4.55)
4	8,379	6.78 (6.49–7.15)	6.48 (6.29–6.92)
5	8,378	10.11 (9.76–10.41)	10.30 (10.05–10.79)
6	8,379	14.09 (13.70–14.57)	15.06 (14.74–15.53)
7	8,379	19.69 (19.26–20.15)	21.28 (20.93–21.87)
8	8,379	29.69 (29.18–30.46)	28.85 (28.41–29.46)
9	8,379	40.79 (40.21–41.29)	39.55 (38.97–40.39)
10	8,378	62.99 (62.19–63.37)	64.85 (63.40–68.60)
Day 2			
1	6,845	0.45 (0.39–0.51)	1.05 (0.97–1.14)
2	6,846	1.60 (1.48–1.72)	2.34 (2.23–2.45)
3	6,846	3.54 (3.30–3.72)	4.28 (4.12–4.42)
4	6,845	5.50 (5.24–5.76)	6.83 (6.64–7.08)
5	6,846	8.72 (8.40–9.00)	10.19 (9.94–10.48)
6	6,846	13.11 (12.66–13.48)	14.68 (14.42–15.05)
7	6,845	19.78 (19.30–20.29)	20.95 (20.64–21.35)
8	6,846	29.83 (29.40–30.30)	28.37 (27.86–28.98)
9	6,846	40.88 (40.37–41.38)	40.08 (39.42–40.77)
10	6,845	65.52 (65.20–65.79)	66.67 (65.98–67.48)
Day 3			
1	5,143	0.70 (0.62–0.80)	1.34 (1.24–1.46)
2	5,144	2.25 (2.08–2.41)	2.84 (2.72–2.96)
3	5,143	3.78 (3.48–3.99)	4.92 (4.78–5.11)
4	5,144	6.19 (5.89–6.55)	7.58 (7.42–7.84)
5	5,143	8.99 (8.67–9.39)	11.11 (10.91–11.40)
6	5,144	14.44 (14.02–14.87)	15.73 (15.47–16.02)
7	5,144	20.95 (20.37–21.42)	22.08 (21.67–22.44)
8	5,143	30.23 (29.69–30.80)	29.99 (29.45–30.52)
9	5,144	44.31 (43.72–44.77)	42.27 (41.70–42.92)
10	5,143	67.38 (67.10–67.72)	69.29 (68.63–70.00)

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Appendix—Observed and predicted pADE incidences by decile of the C-score[a]

Decile, by Day	*n*	Mean Observed pADE Incidence (%, 95 CI)	Mean Predicted pADE Incidence (C-Score, 95% CI)
Day 4			
1	3,849	0.94 (0.81–1.09)	1.71 (1.60–1.84)
2	3,849	2.23 (2.05–2.57)	3.37 (3.23–3.50)
3	3,849	4.75 (4.39–5.09)	5.62 (5.39–5.79)
4	3,849	7.78 (7.43–8.18)	8.49 (8.24–8.72)
5	3,849	11.21 (10.70–11.72)	12.18 (11.83–12.46)
6	3,850	16.10 (15.48–16.65)	16.88 (16.43–17.21)
7	3,849	21.88 (21.17–22.58)	22.85 (22.37–23.29)
8	3,849	30.45 (29.80–31.12)	30.17 (29.64–30.80)
9	3,849	43.59 (43.00–44.25)	42.15 (41.49–42.83)
10	3,849	66.68 (66.22–67.13)	68.65 (67.91–69.29)
Day 5			
1	3,008	1.26 (1.13–1.46)	1.90 (1.76–2.04)
2	3,009	3.24 (2.92–3.52)	3.77 (3.59–3.93)
3	3,009	4.91 (4.65–5.18)	6.29 (6.05–6.57)
4	3,009	7.61 (7.15–8.04)	9.42 (9.08–9.81)
5	3,009	11.95 (11.43–12.63)	13.40 (13.02–13.90)
6	3,009	17.09 (16.35–17.78)	18.42 (17.95–19.04)
7	3,009	25.48 (24.49–26.25)	24.54 (24.00–25.25)
8	3,009	32.72 (31.94–33.77)	32.19 (31.47–33.10)
9	3,009	47.30 (46.49–48.12)	44.93 (44.08–45.77)
10	3,008	69.51 (69.05–69.95)	71.03 (70.34–71.85)

[a]pADE = preventable adverse drug event, C-score = complexity score, CI = confidence interval.